Towards a Benchmark for LLM-based Points of Interest Classification: Open Problems and Preliminary Findings

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Abstract

OpenStreetMap (OSM) Points of Interest (POIs) use inconsistent, crowdsourced place-type labels, making it difficult to evaluate POI classification in a standardized way. In this ongoing work, we develop a 37-category taxonomy and a hybrid approach for mapping OSM labels to these categories. We evaluate POI classification from names alone on 37,586 gold-labeled Manhattan POIs, comparing a fine-tuned transformer with zero-shot and few-shot LLM prompting. The transformer achieves 86.6% accuracy, compared with 18.8% for zero-shot and 55.8% for few-shot prompting. However, 44.5% of POIs share a placeholder name, allowing the model to benefit from memorization; accuracy on real-named POIs is 83.0%. A preliminary analysis of 102,936 Paris POIs shows that 94.9% of place-type labels can be mapped using the same taxonomy. These early results highlight a significant data-quality issue and suggest that the proposed taxonomy and evaluation approach could support more reliable cross-city studies of POI classification.

Keywords

Points of Interest, Large Language Models, benchmark construction

1. Introduction

Every point on a map that represents a real place, such as a restaurant, a shop, a bank, or a park, is called a Point of Interest (POI). Knowing what *kind* of place each POI is matters for many practical tasks: understanding where and why people move through a city, recommending places to visit, or simply building a clean, organized map. The largest open source of this kind of data is OpenStreetMap[1] (OSM), a map that anyone in the world can edit, much like Wikipedia. This openness is also its weakness, because anyone can describe a place however they like: the same type of place ends up labeled in many different, inconsistent ways across the map. For example, a cafe like "Starbucks" might be tagged by one contributor as a *cafe*, by another as a *fast food*, and by a third simply as a *beverage*. For practical purposes, these should all count as the same broader category, for instance *food and beverage*, but OSM has no built-in notion of this broader grouping, nor a fixed list of allowed labels, and to our knowledge, there is no single, agreed-upon, finite list of place categories that all this data cleanly fits into.

Large Language Models (LLMs) offer an inviting way to address this problem by simply showing the model a place's name and asking which category, out of a fixed list, it belongs to. If this worked reliably, it would make it far easier to organize POI data consistently across any city, and it would be useful for downstream tasks such as POI recommendation [1, 2], next-POI prediction [3], urban region representation [4], and other applications that rely on having a consistent, large-scale representation of what a place is [5, 6]. The problem is that, to the best of our knowledge, this has not yet been tested in a rigorous way. There is currently a scarcity of standard benchmarks: no agreed-upon test, with a proper answer key, for measuring whether an LLM can correctly classify POIs, nor is it known whether an LLM performs consistently well across different cities or shows hidden biases toward the places it has seen more of. In the literature, [7] proposes a solution to a related challenge, aligning

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[1]https://www.openstreetmap.org

OSM place-type labels with the Foursquare[2] category system using pretrained sentence-transformer embeddings refined by a generative LLM, producing a large-scale, hierarchical benchmark against Foursquare's six-level hierarchy. Our work differs in two aspects: rather than aligning OSM's existing labels to a large commercial taxonomy, we build a small, finite, flat taxonomy from scratch together with a domain expert; and rather than only aligning structured tags, we specifically study classification from a POI's *name* alone.

This paper reports on our ongoing work to build this kind of benchmark and describes the concrete difficulties we encounter along the way. The methodology we investigate proceeds in three steps: first, cleaning and consolidating the raw labels attached to real POIs into a small, finite, human-validated set of categories (Section 3.1); second, training and prompting different classifiers to predict this category directly from a POI's name alone (Section 3.2); and third, evaluating whether this name-based classification is both generalizable across different cities and reproducible by other researchers (Section 3.3). We compare three well-established classification techniques under this design. Fine-tuned transformer classifiers, following the BERT architecture [8], serve as our supervised baseline. In zero-shot and few-shot prompting models [9], we ask a pretrained model to perform a task directly from natural-language instructions, optionally including a small number of labeled examples in the prompt, with no weight updates at all. Finally, for the planned agreement and error-analysis layers of our evaluation, we adopt the LLM-as-judge paradigm [10], in which a language model is used to assess the quality of the output of another model. Our preliminary results (Section 3) show that a frozen LLM only gets close to the fine-tuned classifier's accuracy when it is given a few labeled examples per category; without any examples, its accuracy is much lower. We are working on better ways to select examples to close this gap.

2. Problem Statement and Challenges

Building a fair and careful benchmark for POI classification from name alone, as introduced above, raises four concrete problems.

(1) Lack of an agreed-upon list of categories. Because OSM's place-type labels are crowdsourced, the categories assigned to real-world places are sometimes inconsistent, contradictory, or noisy, as in the Starbucks example discussed in Section 1. To make our pipeline reproducible, and its results comparable across cities, we first need a small, finite, agreed-upon list of categories that this noisy information can be consistently mapped onto. We describe how we build one in Section 3.

(2) Lack of an existing set of correctly-labeled examples. Even with an agreed list of categories in hand, we lack a dataset that already tells us the correct category for real-world places. This has to be constructed by hand, place by place and label by label, which turns out to be a far slower and more detailed task than it appears at first.

(3) Missing names can bias evaluation results. We find that a large share of places in our data have no usable name at all; in the source data, this is represented by a single generic filler value used in place of a missing field. Because many different places share this exact same filler value, a supervised classifier can achieve high accuracy by learning which category the filler value is usually associated with, rather than by learning to interpret real names. A prompt-based classifier, which cannot exploit this shortcut, does not benefit in the same way, and can therefore appear weaker in a raw comparison, even though in both cases no genuine name-based information was actually available. We consider this a general risk for any name-based POI classification benchmark, and describe how we identify and account for it in Section 3.2.

(4) Unknown consistency of LLM-based classification across cities. A language model may classify places more accurately in a city, or a language it encounters often during training, and less accurately somewhere less familiar to it. Whether this kind of geographic inconsistency actually happens, and how large it is, remains an open question that, to our knowledge, has never been properly tested.

[2] https://docs.foursquare.com/data-products/docs/categories

Table 1
Name-only classification accuracy (%), Manhattan test set (n=7,518), split by name availability.

Classifier	All POIs	Placeholder	Real-named
BERT (fine-tuned)	86.6	91.0	83.0
Zero-shot LLM	18.8	10.1	25.7
Few-shot LLM	55.8	77.4	38.4

3. Preliminary Investigation

We report here the current status of our ongoing work, focusing on three step methodology and qualitative findings till now.

3.1. Step 1: Cleaning and Consolidating Place Categories

Working with a domain expert, we adopted a finite taxonomy of 37 place categories and cross-referenced it against the distinct place-type labels observed in an OSM extract of Manhattan[3]. As the example illustrated in Section 1, these raw labels are a lot and weakly structured: out of 115 distinct labels observed, 60 were resolved through the domain expert's existing reference list of label-to-category mappings (direct matches). The remaining 55 labels were not covered by this reference list and were resolved through a hybrid protocol: an open-weight LLM (Qwen2.5-7B-Instruct [11], run locally with no fine-tuning) proposed a category for each unresolved label from the closed 37-category vocabulary; every proposal was then manually reviewed, and a small residual of genuinely ambiguous cases was escalated back to the domain expert for a final decision. This process yielded a complete label-to-category lookup table, in which every entry retains a provenance label, **direct match**, **LLM-resolved**, or **expert-confirmed**, recording how it was decided. Applying this lookup to the full Manhattan POI extract produced a gold-labeled collection of 37, 586 POIs, which we further split into training and test partitions using sample-wise sampling, so that every category is represented in both partitions, a necessary condition for fairly evaluating a supervised classifier. Tracking in each individual entry, rather than just providing a raw lookup table, is very important for the testability of this resource.

3.2. Step 2: Classification from POI Name Alone

Classifier comparison. We compare three classifiers, all restricted to predicting category from POI name alone, a fine-tuned supervised transformer (BERT-base), a zero-shot prompted LLM, and a few-shot prompted LLM, as introduced in Section 1. Table 1 reports their accuracy overall and in each split by whether the POI had a genuine name on the Manhattan test set.

The fine-tuned baseline achieves the highest accuracy on the POIs with a real name, specifically, and it outperforms the other two prompted models, indicating an advantage when labeled training data is available. However, part of its overall advantage turned out to be due to a risk in the quality of the data that we describe in the next Section 3.2. The few-shot LLM improves over zero-shot prompting on both subsets, suggesting that a small number of labeled examples helps an LLM to adapt to a custom taxonomy, even though we note that our current few-shot example set left 14 out of the 37 categories, mostly infrastructure and land-related categories that are more under-represented among genuinely-named training POIs, without any real-named example at all, which we expect to address in future work.

A data-quality risk for name-based evaluation. While analyzing these results, we found that 44.5% of POIs in our Manhattan dataset, across both the training and test partitions, shared a single placeholder name introduced upstream whenever a POI's true name was absent from the source data. The underlying fact that many POIs (mostly street furniture such as benches, waste baskets, and

[3]Extracted via Geofabrik: https://www.geofabrik.de/

surveillance cameras) genuinely lack a name is not itself an error; the artifact lies in substituting one identical string for every such case. We observed that this allows a supervised classifier to partially achieve high accuracy through memorization rather than genuine name understanding. Our fine-tuned baseline's accuracy on placeholder rows (91.0%) is statistically indistinguishable from a naive majority-class baseline computed on the same subset, while its accuracy on genuinely-named POIs (83.0%) reflects real name understanding. Prompt-based methods cannot exploit this shortcut and initially appear disproportionately weaker as a result. However, we observed an interesting secondary effect: a few-shot model's labeled examples never included the placeholder string itself, yet the model still correctly classified most placeholder rows. We attribute this to the model recognizing the word "infrastructure" within the placeholder text and associating it with the same category as its few labeled examples of unrelated utility services (e.g., postal and public Wi-Fi infrastructure). We found this to be a fundamentally different mechanism from the supervised model's behavior. Rather than memorizing a single, repeated string, the few-shot model appears to generalize from the *meaning* of a small number of examples to new text that shares related vocabulary, even without any exact overlap in wording.

3.3. Step 3: Evaluating Reproducibility

To check this we applied the same tag-resolution lookup from Section3.1 to the Paris region (Île-de-France OSM extract from Geofabrik), without any modification. Around 102,936 POIs in Paris were found, with 118 distinct place labels among them. The majority of labels were already resolvable as: 52.5% through a direct match already present in our list, and a further 42.4% through the same zero-shot LLM plus manual-review protocol used for Manhattan, together covering 112 of the 118 labels (94.9%). Of the remaining six labels, two were new to Paris and were resolved by the domain expert's reference list, while the four uncovered labels were *town hall*, *water well*, *wayside cross*, and *wayside shrine*. This suggests that the underlying taxonomy and resolution methodology may generalize to a new city with modest additional effort, though a full evaluation of whether classification *quality*, and not only label *coverage*, transfers as well is ongoing work.

We are also adding two more checks: how well classifiers agree with the labels the domain expert reviewed directly, and whether an LLM judge sees the errors as reasonable near-misses or real errors. We leave these results, for a future analysis, where we also plan to release the taxonomy, the method, and the evaluation protocol as a reusable benchmark.

4. Conclusion and future work

Overall, we are moving towards a citable, reusable benchmark for LLM-based POI classification, built on a finite, domain-validated taxonomy. Our preliminary results show that a fine-tuned classifier outperforms zero-shot and few-shot LLM prompting on name-based classification of POIs, while also exposing a risk: missing POI names can collapse into a single placeholder value that silently distorts such comparisons if left unaddressed. This risk, along with the open question of whether LLM-based classification remains consistent across cities, is relevant to any similar benchmark. Our immediate next steps are to complete the three-layer validation described above, consisting of exact-match comparison, agreement analysis against the labels the domain expert directly validated, and an LLM-as-judge assessment of near-miss errors; to extend the cross-city evaluation to test whether classification quality, not just label coverage, transfers to a second city, with explicit attention to geographic consistency; and to formalize and, where possible, publicly release the gold-labeled benchmark dataset and evaluation protocol, to support reproducible comparison by other researchers working on LLM-based POI classification .

Declaration on Generative AI

During the preparation of this work, the authors used Claude for grammar checking and rephrasing. The content has been reviewed and we take full responsibility for it.

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